

Physical Parameter Learning in Complex Initial Condition

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Motivation

Limitations of traditional system identification:

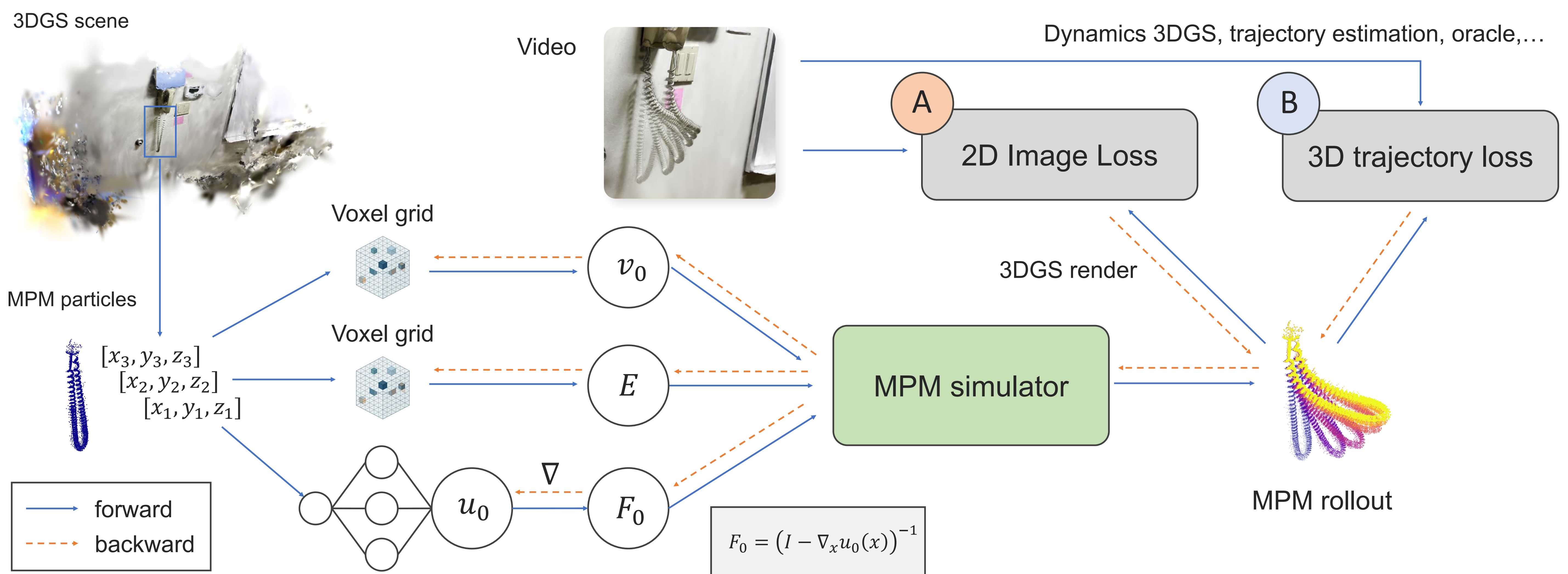
- Assume object to be rest
- Homogeneous parameter learning

Can we deal with more complex setup?

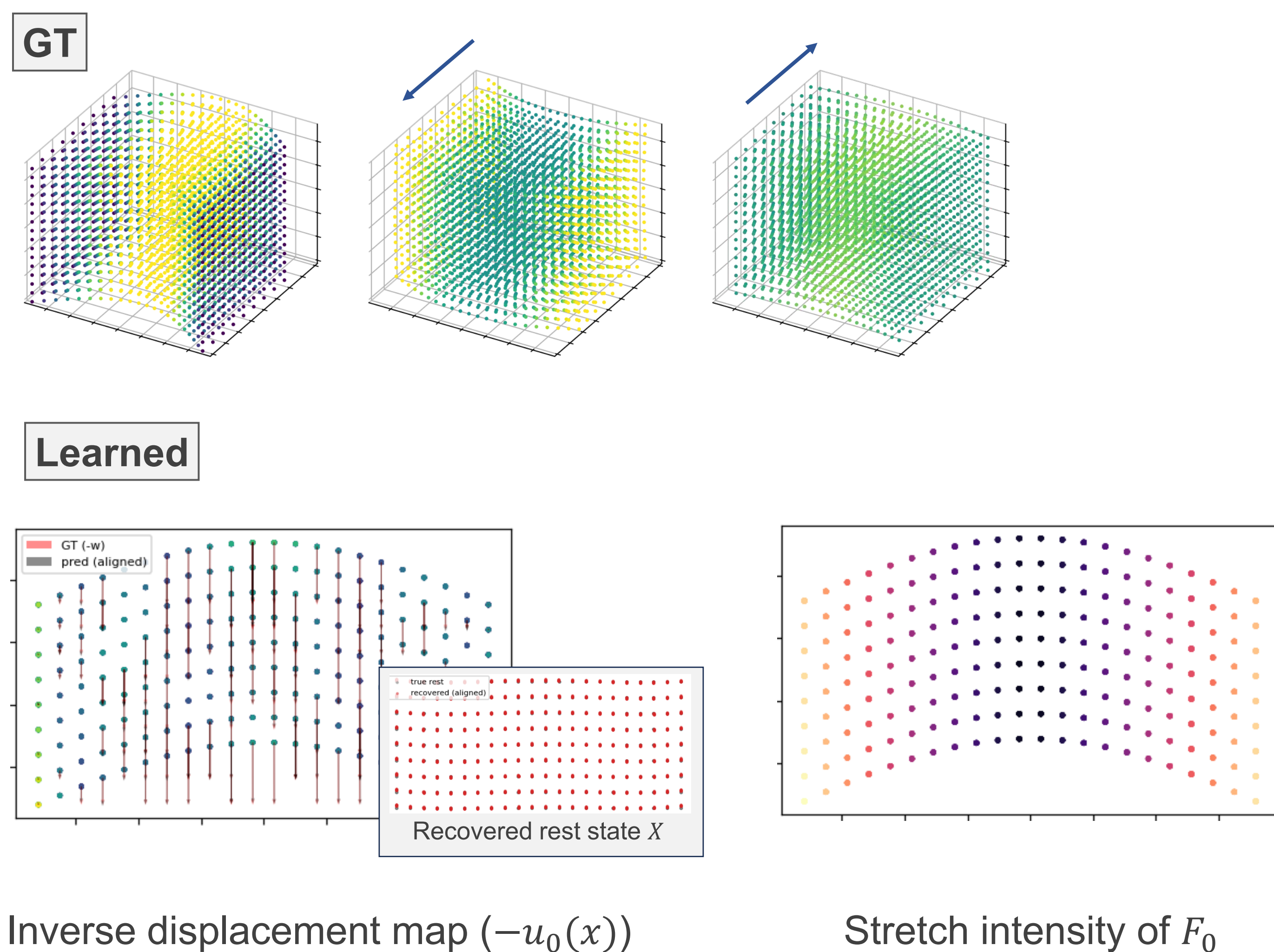
Takeaways

- First learnable deformation gradient with differentiable MPM
- Support inhomogeneous initial velocity and material
- Enhanced robustness with multi observation training

Method

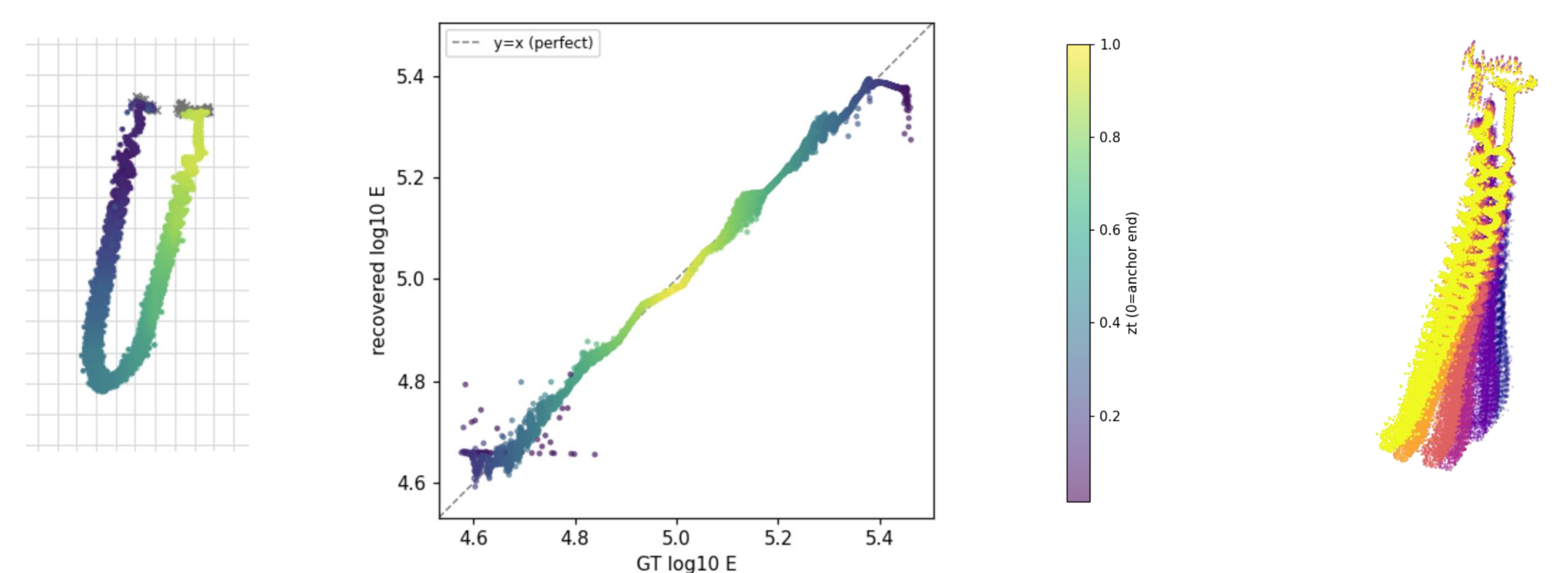


Learning initial deformed gradient

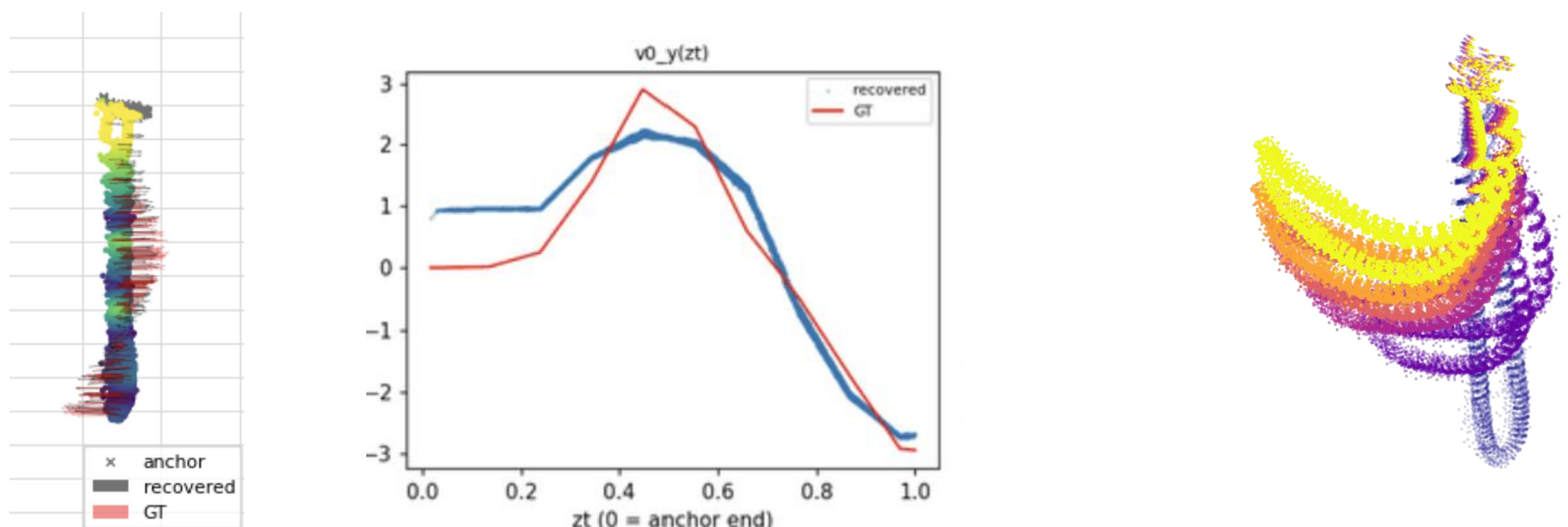


Inhomogeneous parameter fields

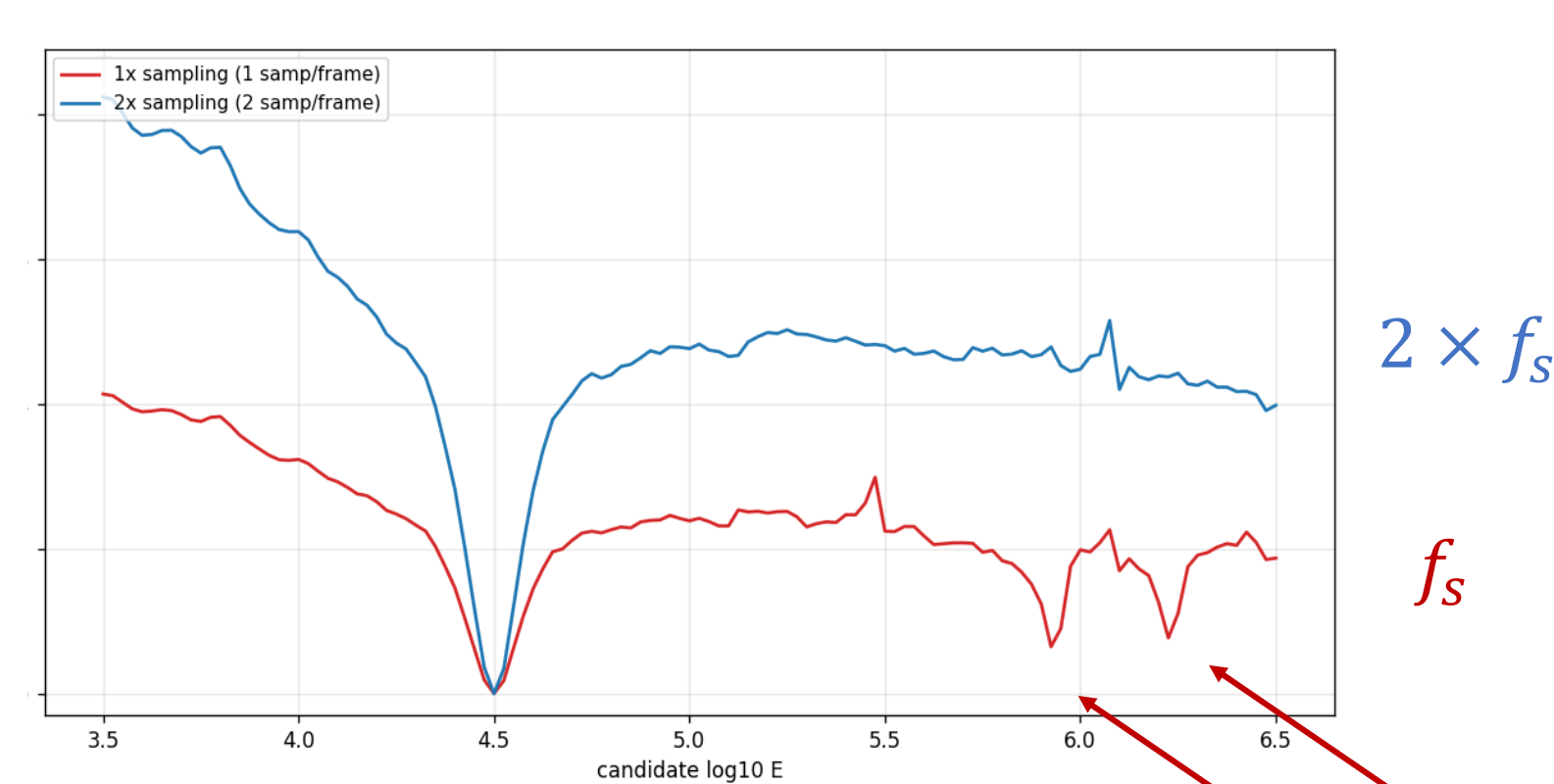
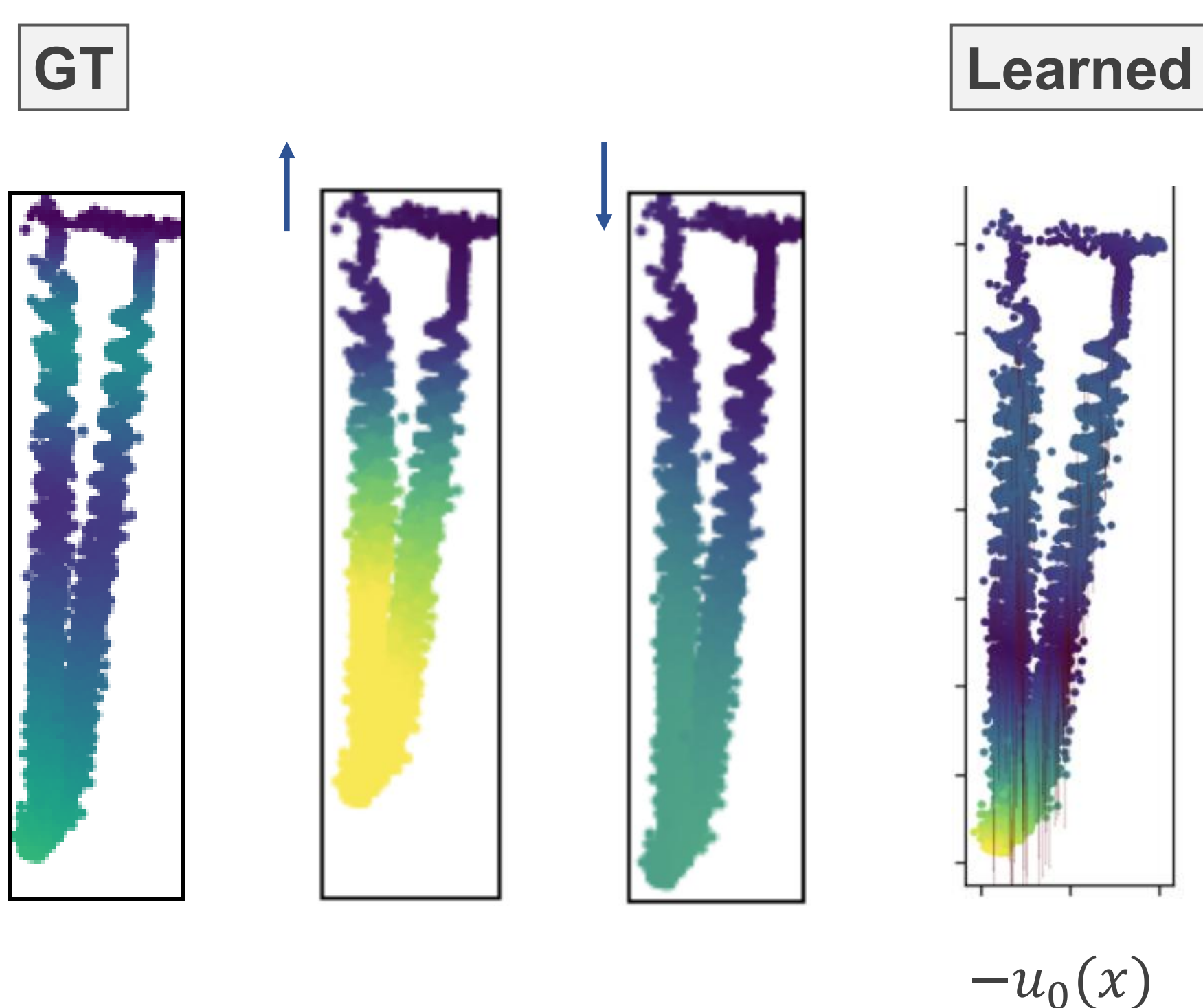
- **Learned** Young's modulus (E) field



- **Learned** initial velocity (v_0) field



Local Minima Problem: Oscillation and Sampling



- $\omega \propto \sqrt{E}$; Phase shift $\Rightarrow$ plateau
- Aliasing $\Rightarrow$ Local minima

